



CHAOTIC DYNAMICS AND FRACTAL SPACE IN BIOCHEMISTRY: SIMPLICITY UNDERLIES COMPLEXITY

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In this work we attempt to analyze the coupling between the dynamics of biochemical reactions (especially chaotic dynamics), and the geometry of cytoarchitecture (especially fractal ultrastructure), because of its importance and consequences for the ultradian dynamic behaviour of cells. Fractal geometry in intracellular macromolecular assemblies suggests that chaotic dynamics occur during their organization. Non-linear interactions in and between spatial and temporal domains and over wide ranges of scales underlie the emergent properties of complex biological systems.

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INTRODUCTION

The ability of biological systems, either unicellular or multicellular, to exhibit rhythmic behaviour in the ultradian domain is a fundamental property because of its potential role as an inducer of coherence, e.g. entrainment and co-ordination of intracellular functions (Lloyd, 1992). In systems exhibiting chaos, many possible motions are present simultaneously. In fact, since the dynamics of a chaotic system trace a strange attractor in the phase space, in principle a great number of unstable limit cycles are embedded therein; each of these is characterized by a distinct number of oscillations per period (Peng *et al.*, 1991; Shinbrot *et al.*, 1993). Biological systems exhibiting chaotic dynamics need only small perturbations of their parameters in order to select stable periodic outputs (Lloyd and Lloyd, 1993, 1995). This characteristic facilitates dynamic motion of the system between attractors, i.e. *homeodynamics* (Lloyd, Aon and Cortassa, 2000, submitted).

It has been proposed that the ultradian clock has timing functions providing a time base for intracellular co-ordination (Lloyd, 1992). The intracellular spatio-temporal co-ordination of gene

expression, energetics, and intermediary metabolism, is at the heart of a cell's growth, proliferation, differentiation, or functional adaptation to environmental changes (Aon and Cortassa, 1997). Key to the spatio-temporal organization of cellular dynamics is the geometry and dynamics of its cytostructure. It is becoming increasingly evident that the understanding of how a cell interprets signals from a developmental field or reacts to environmental stress is crucially dependent on the spatio-temporal organization of its cytostructure (Aon and Cortassa, 1997; Aon *et al.*, 2000).

Both geometrically and dynamically, biological systems are intricate (Lloyd and Kippert, 1993; Aon and Cortassa, 1997). Concerning geometry, the natural world cannot be described in terms of the familiar geometry of lines, triangles, squares and circles. Instead, mountains, clouds, and coastlines are fractal structures that always look the same as portions of them are enlarged (Mandelbrot, 1983). Fractal forms are heterogeneous, self-similar, and without a well-defined scale of length. Fractal structures have irregularity and redundancy, and as a consequence are robust to injury.

Chaos occurs in non-linear, thermodynamically open systems performing close to, but not under, equilibrium conditions. A chaotic attractor is a

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geometrically fractal object. When the trajectories followed by the dynamics of a system settle or approach a restricted region of the phase space, that region is called an attractor (Aon and Cortassa, 1997; Lloyd, Aon and Cortassa, 2000, submitted). Often the link between chaos and fractals has been established due to the fractal geometrical structure of variable trajectories for strange attractors. It is now clear that non-linear systems can exhibit strange attractors that may be quantitatively described by fractal geometry (Glass and Mackey, 1988). However, in cellular dynamics the link between chaos and fractals can be more subtle and interesting. Indeed, the many behavioural possibilities exhibited by dynamic systems correspond to a set of attractors, that represent the integrated dynamic behaviour of the variables in the system, as they mutually influence one another. In this sense, the dynamic supramolecular organization of components of the cytoskeleton gives rise to intricate fractal geometries in cells, and sophisticated spatio-temporal behaviour of cellular biochemistry and energetics (Aon and Cortassa, 1997). In the present work we intend to analyze possible relationships between chaotic dynamics and the fractal geometry of the cytoarchitecture.

THE GROUND SUBSTANCE OF CELLULAR CYTOPLASM

Over the past twenty years the complexity of the cytoplasm has become increasingly evident, and ideas about the unexpected ways in which the behaviour of processes occurring therein have blossomed. The structured nature of the milieu may often provide directional ordering (vectorial metabolism, channeling). The heterogeneity of solvent states and properties in space and time may accelerate or hinder reaction kinetics. The fractal nature of cytoplasm (Aon and Cortassa, 1994; Forgacs, 1995) also suggests novel and unexpectedly organized physiology (see below).

The 'ground substance' of the cytoplasm consists of a complex dynamic network of polymeric elements, the cytomatrix, which includes an array of actin and tubulin filaments (Clegg, 1984; Schliwa and Van Blerkom, 1981), which together with their many associated structural proteins, provides the molecular latticework for the positioning and distribution of sub-cellular organelles and vesicles (Knull and Minton, 1996). The framework of the system is itself continuously being dissolved and reformed. The rapid translocation of membraneous structural elements is powered by ATP-driven motor proteins.

The dynamic instability of microtubular structures is modulated by their associated proteins. The regulation of this interplay is by microtubule affinity regulating kinases, and it is via this system that morphogenesis (cell polarity, shape and motility) is elicited by signal transduction (Drewes *et al.*, 1998). Turnover of microtubules is increased as much as 18-fold during mitosis (Saxton *et al.*, 1984).

The supramolecular structure of ovomucin serves as a model system for the study of the complex rheological properties and dynamic organization of the cytosol (Rabouille *et al.*, 1992; Aon and Cortassa, 1997). Slow drying of an ovomucin gel/salt system gave dendritic-like fractal patterns when the protein:salt ratio was optimum (within the range 0.75–1.25) and the calculated value of the fractal dimension (1.79) was similar to that described for percolation clusters (Rabouille *et al.*, 1992; Aon and Cortassa, 1997). It was shown that various other biopolymers (fetuoin, albumin or starch), behave similarly. It appears that salt percolates through the macromolecular lattices and drying gives precipitation in fractal clusters (Fig. 1).

Superimposed on the cytomatrix system of enzymes is the associated network of 'cytosolic' enzymes (e.g. those of glycolysis). Free and cytomatrix-bound enzymes exchange rapidly. In the bound state the enzymes may form supramolecular aggregates in configurations that facilitate and accelerate metabolic fluxes through spatially organized routes (metabolic channeling; Welch, 1984).

For an enzyme it has been suggested that, as well as the specific structure innate in substrate binding sites, allosteric control sites, and the array of native configurational states, more global statistical patterns might be important (Pfeifer *et al.*, 1985). Thus the theme of invariance under dilation (that scaling up of any part of the system reproduces the system itself), has been shown for the lysozyme molecule. Fractal surface dimension ($D=2.17 \pm 0.02$) in this case controls its biological activity. The rate of substrate trapping by the active site is largest when the protein surface has a dimension larger than two, as seen by water (taken as a conveniently-sized yardstick). A surface dimension >2 in effect accelerates diffusion to the surface, but slows it down on the surface. The optimum for substrate capture was calculated to be close to $D=2.2$.

INTRACELLULAR EFFECTORS OF CYTOSKELETON DYNAMICS

Main cytoskeletal proteins such as microtubules (MT), actin and intermediate filaments represent

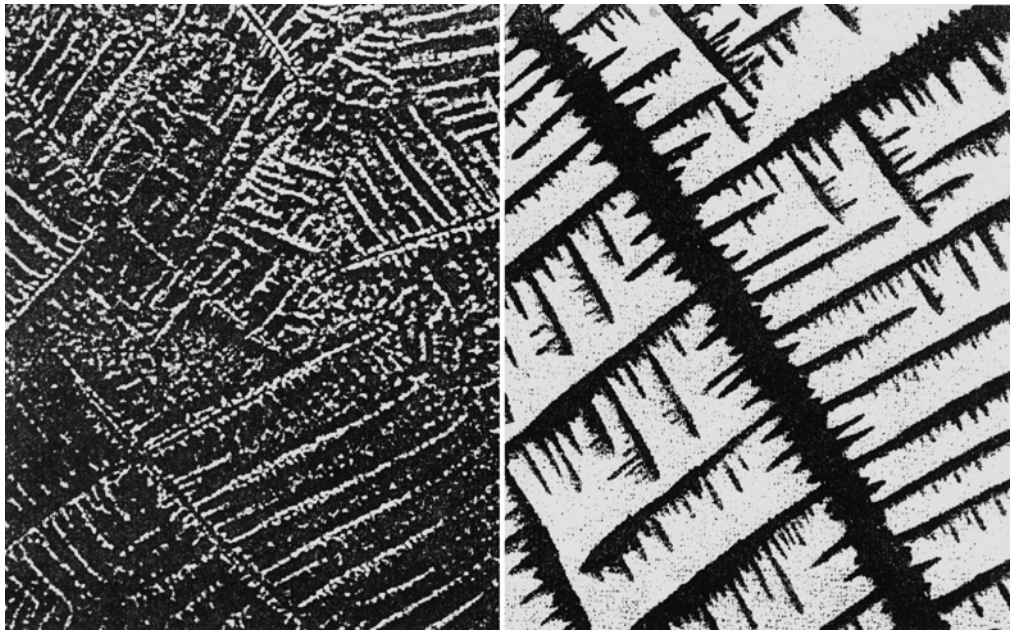


Fig. 1. Fractal patterns in biomolecules. Dendritic-like fractal patterns in an ovomucin gel/salt system. Dendritic-like patterns observed by light microscopy ($\times 50$; left panel); electron microscopy of dendritic-like patterns ($\times 1200$; right panel). Notice that a self-similar structure is apparent when both panels are compared just by eye and as expected they exhibit the same fractal dimension $D=1.79$ ($r=0.996$) (from Rabouille *et al.*, 1992).

an enormous protein surface in the cells. Particularly, MT present a protein surface in the cell of about 1000 nm^2 in the interphase array of an average proliferating fibroblast (Gundersen and Cook, 1999). Several proteins bind to this interphase, its binding depending upon protein phosphorylation status (Pedrotti *et al.*, 1996) and affecting MT dynamics. The relevance of intracellular effectors of cytoskeleton dynamic organization is direct, given by the fact that its intrinsic dynamics may, in turn, affect the dynamics of enzymatic reactions or of signalling pathways (Aon and Cortassa, 1997, and Aon *et al.*, 2000, for reviews; Gundersen and Cook, 1999).

Indeed, evidence has been reported that indicates ultrastructural effects on MT triggered by enzyme binding. *In vitro* data showed that MT ultrastructure was modulated by glyceraldehyde-3-phosphate dehydrogenase and phosphofructokinase, brain- and muscle-specific pyruvate kinase M1 isoform, but not aldolase (Vertessy *et al.*, 1997; Vertessy *et al.*, 1999).

Existing evidence reveals signalling molecules that interact with MT and display multiple effects on signalling pathways of drugs that breakdown (e.g. nocodazole, vinblastine), or hyperstabilize MT (e.g. taxol and taxotere), indicating that these cytoskeleton polymers are likely to be critical to the spatial organization of signal transduction (Gundersen and Cook, 1999). Sequestering and

release, delivery and scaffolding of signalling molecules have been proposed as transduction mechanisms mediated by MT. Recent studies show that changes in MT dynamics and organization occur rapidly in response to activation of signalling pathways as changes in (i) MT dynamic instability; (ii) MT stability; (iii) overall MT organization (Gundersen and Cook, 1999).

THE DYNAMICS OF METABOLIC AND SIGNALLING NETWORKS AND THE FRACTAL NATURE OF CYTOPLASM

Since it was suggested that the cellular cytoplasm can be described as a percolation cluster, a sort of random fractal (Aon and Cortassa, 1994, 1997; Forgacs, 1995; Aon *et al.*, 2000) it became timely to know what would be the consequences for the dynamics of intracellular metabolic and signalling networks.

A good example of percolation is the growth of clusters on an infinite square lattice (Feder, 1988; Stauffer and Aharony, 1994; Dewey, 1997). A cluster is defined as a group of particles that forms contiguous nearest neighbours. At a critical density of sites, all the clusters will be connected into one infinite cluster. This is the percolation threshold, p_c . When a cluster spans a lattice at p_c , a percolating cluster is formed. The resulting percolation cluster is a fractal. Thus, percolation implies fractality.

The rheological properties (Srisvastava and Krishnamoorthy, 1997) of the ground matrix as a heterogeneous highly structured milieu for the enzymes associated within it may be fractally organized as an ensemble of pores or sites in a lattice (Aon and Cortassa, 1994; Forgacs, 1995). Percolation through such a lattice above a threshold gives large-scale connectivity (Feder, 1988; Aon and Cortassa, 1994; Forgacs, 1995; Aon *et al.*, 2000). Thus a percolation cluster can undergo a transition from local behaviour to coherence over the whole of the cytoplasm. Thus metabolites, hormones, second messengers and ions percolate through this fractal system so as to connect local processes with those occurring at remote sites (Aon and Cortassa, 1994, 1997).

Particularly important are the consequences for catalysis and dynamics of biochemical reactions in heterogeneous fractal media. Enhancement of catalysis in fractal media has been predicted: (i) by an increase in reactivity per unit surface area due to segregation of reactants (Kopelman, 1988), and (ii) because a volume may fractally evolve into an area by fractal folding, and the probability of a random walker finding a target is higher in two rather than three dimensions (McCloskey and Poo, 1986). Indeed, reduction of dimensionality is a main aspect of catalysis since lower topological dimensions (2D) increase the probability of encounter between enzyme and substrate, or effector and target, with respect to higher dimensions (3D). In fact, the mean diffusion time τ for a molecule to reach a specific target site in a space of diameter L is proportional to L^2 for one and two dimensions, and is proportional to L^3 in three dimensions. Thus, a biological system can greatly enhance the targetting of a small molecule to an enzyme or receptor by reduction of dimensionality (Adam and Delbrück, 1968; Dewey, 1997).

A prediction of the fractal model of cytoplasm is that a cell, by regulating the geometry or architecture of its cytoskeleton, in turn regulates the level of its percolation threshold, p_c . Different p_c levels will determine when the local level of a messenger or the product of an enzymatic reaction may extend to the whole cellular field if it spreads in a domain belonging to the spanning cluster (Aon and Cortassa, 1997; Aon *et al.*, 2000).

COUPLING THE DYNAMICS OF CHANGES IN CYTOSKELETON ORGANIZATION AND OF ENZYMATIC REACTIONS

A greater amount of work has been done on protein–cytoskeleton interactions than on the func-

tional consequences of these interactions. By functional consequences we mean changes in fluxes through enzyme-catalyzed steps, as well as in signal transduction pathways.

Changes in the dynamics of tubulin assembly-disassembly may entrain the dynamics of enzymatic reactions (Cortassa and Aon, 1996; Aon *et al.*, 1996; Aon and Cortassa, 1997). The latter highlights not only the importance of knowing the intracellular effectors of cytoskeleton dynamics, but also the effects on the dynamics of enzymatic reactions and targeting of intracellular signalling pathways. Differential modulation of the activities of glycolytic enzymes occurs depending on whether the reactions are measured in the presence of polymerized or non-polymerized microtubular protein (MTP) (Cortassa *et al.*, 1994).

It was suggested that two important characteristics of microtubular networks might make them likely candidates for the coherent synchronization of intracellular dynamics (Cortassa *et al.*, 1994; Aon and Cortassa, 1994, 1997): (i) their non-linear dynamics which at some non-equilibrium state might give rise to self-organization and macroscopic structuring, and (ii) their fractal or self-similar nature over a range of spatial length scales (Rabouille *et al.*, 1992; Aon and Cortassa, 1997).

Intracellular signalling is arranged in a large number of components comprising their interactions and spatial relationships as well (Weng *et al.*, 1999). Networking results in several emerging properties that the individual pathways do not have, in this way supplying an additional store of information within intracellular biochemical reactions of signalling pathways. Although mutations or altered gene expression can result in persistent activation of protein kinases, connections between pre-existing pathways may also result in persistently activated protein kinases capable of eliciting biological effects (Bhalla and Iyengar, 1999).

In systems where two signalling pathways interact through a feedback loop, the amplitude and duration of the extracellular signal may determine the sustained activation of the system (Bhalla and Iyengar, 1999; Weng *et al.*, 1999).

FRACTAL GEOMETRY IMPLIES CHAOTIC DYNAMICS

The fractal structure of the highly branched respiratory and circulatory systems is taken to be a consequence of the slow evolutionary and developmental dynamics of the processes leading to their elaboration (Goldberger *et al.*, 1990). In general,

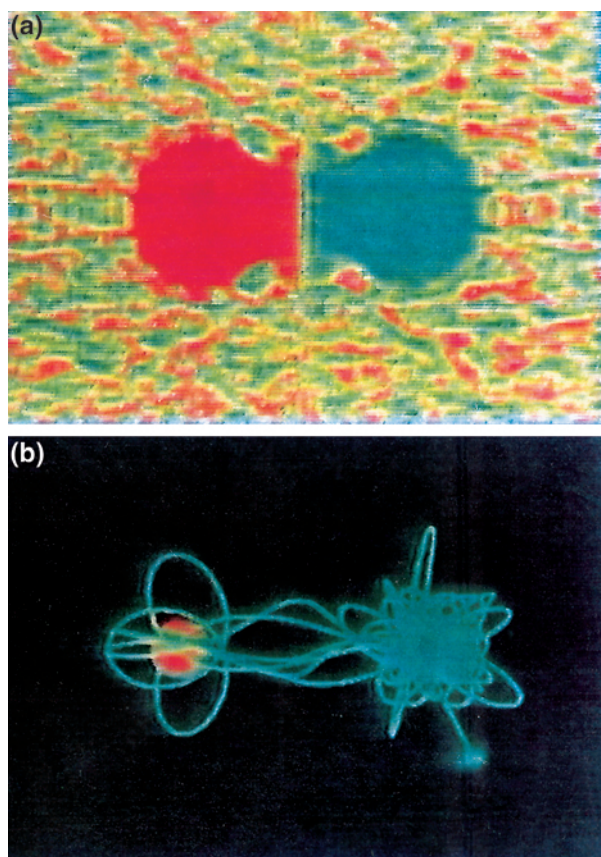


Fig. 2. S mechanical chaotic system consisting of two magnets A (green) and B (red) and suspended magnetic pendulum. Tracing back the trajectories from either of the eventual equilibrium positions (on A or B) gives a fractal pattern for the initial points of release.

the concept of fractality can pertain to both space and time (Yates, 1992). Chaotic dynamics may generate quite elaborated (chaotic) patterns (Stewart and Golubitsky, 1992). The generation of fractal structures may require the operation of chaotic dynamics. Chaotic processes acting on the atmosphere or on the landscape lead to the production of fractal objects such as clouds, coastlines and eroded rock surfaces. An example of the relationship between chaotic dynamics and fractal structure is shown in Figure 2.

Time series analysis can help determine the relative contributions of deterministic and stochastic processes to biological processes (Lloyd and Lloyd, 1995). A log-log plot of power (i.e. amplitude), versus frequency for a periodic process gives a line of negative slope^m (West, 1987). White noise gives a slope of zero; Brownian noise (diffusion) -2 , and Mandelbrot noise approximately -1 . This latter characteristic of $1/f$ noise is diagnostic (although not conclusively so), of fractal time, indicative of a scale-invariant distribution of re-

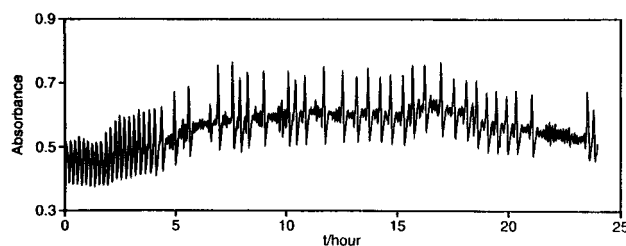


Fig. 3. Readout of $A_{340\text{ nm}}$ relative to that at 400 nm. The CSTR was continuously supplied with 100 mM glucose and a yeast extract with 9.9 mg/ml protein. (ATP)=3.1 mM (NADH)=0.5 mM and $(\text{Mg}^{2+})=10$ mM. The two inflows had the same rate and the total specific flow rate was initially 2.1×10^{-2} /min, decreased to 1.3×10^{-2} /min, and then to 1.2×10^{-2} /min. Simple periodic behaviour at the higher flow rates, gives way at the lowest flow rate to chaotic output (Nielsen *et al.*, 1998).

laxation times. Examples of scale-dependent scatter in physiological or biochemical systems are provided by analysis of capillary blood flow fluctuations (Bassingthwaighe, 1998), and channel openings and closings in patch-clamped lens epithelial cells (Liebovitch *et al.*, 1987). Fractal models gave the better fits in both cases than those with mono- or multi-exponential rate constants. The branching of the dendritic tree of many neurons is fractal and it has been suggested that it may be related to chaos in the nervous system (Goldberger *et al.*, 1985).

OBSERVATIONS OF CHAOTIC DYNAMICS IN BIOCHEMISTRY

The identification of chaotic dynamics in biochemical systems is not straightforward (Yates, 1992; Lloyd and Lloyd, 1995); in order to identify a chaotic attractor, as many as 10,000 data points are required. Whereas this requirement is not a problem for model systems, it places stringent conditions on experimental results. Methods that provide continuous readout are distinctly advantageous.

The first demonstration of chaos in an experimental biochemical system was that with a peroxidase reaction *in vitro* (Olsen and Degn, 1977). With the same enzyme, a period-doubling cascade leading to chaos has been shown in a pseudo-open (Geest *et al.*, 1992), and more recently in a truly open system (Hauck and Schneider, 1994). In a cell-free extract of yeast, glycolytic chaos has been induced by time-dependent forcing with a periodic input of glucose (Markus *et al.*, 1985), and chaos has also been demonstrated (Fig. 3) in

a continuously stirred tank reactor with simultaneous inflows of extract and glucose (Nielsen *et al.*, 1997, 1998).

Chaotic dynamics often results from the interaction between oscillators. Such interactions on longer time scales (in the ultradian, cell division cycle or circadian time domains), may be expected to give rise to self-similarities and chaos (Lloyd and Lloyd, 1993). Control of chaos may provide a tuneable multi-oscillator for the generation of rhythms in different time domains from a single source (Lloyd and Lloyd, 1993). To control chaos, one of the many periodic orbits contained in the attractor is selected and stabilized by periodic perturbation using a feedback control circuit with delay (Meucci *et al.*, 1998; Moss, 1994). Thereby one of a range of periodic outputs can be selected and generated from a single device. We suggest that the complex time structure of living systems may arise in this way (Lloyd and Lloyd, 1993).

CONCLUDING REMARKS

The creation of fractal structures in the physical/biological world may require the operation of chaotic dynamics. The common occurrence of fractal geometries in biological morphologies suggests, but does not prove, that chaotic dynamics are very widespread in nature. Nevertheless, in patchy environments which can be regarded as dynamical functional subsystems (below the percolation threshold), in the cellular cytoplasm (Kauffman, 1989; Aon and Cortassa, 1997), rigidly deterministic unstable dynamics at the local level together with limited diffusion toward adjacent patches, may give rise to an amazing range of complex spatio-temporal patterns such as spiral waves or spatially chaotic variation (Hassell *et al.*, 1991). We note, however, that not always does a chaotic spatial pattern necessarily imply chaotic dynamics, since deterministic systems whose dynamics may become locally unstable and diffusively spread to adjacent patches may give rise to spatio-temporal chaos or complex spatial patterns such as spiral waves or 'crystal lattices' (Hassell *et al.*, 1991; Cross and Hogenberg, 1994). Furthermore, chemical chaos in the Belousov-Zhabotinskii reaction may develop from the interaction of propagating waves and stationary dissipative structures (Agladze *et al.*, 1984).

We have elsewhere suggested that under *homeodynamic* conditions, living systems spatio-temporally co-ordinate their functioning by essentially top-bottom or bottom-up mechanisms.

The former are represented by circadian and ultradian rhythms with clock characteristics, whereas the latter emerge from the intrinsic, autonomous dynamics of the integrated mass-energy-information carrying networks that represent living systems (Lloyd, Aon and Cortassa, 2000, submitted). Thus the apparent intricate fractal geometry of cells (based on the iteration of a simple motif), and its potential as a source for sophisticated (chaotic) dynamics, along with that of metabolic and signalling networks, may be the crossroads of the globally-organized complexity generated by the cross-talk between the two opposing (i.e. top-bottom or bottom-up mechanisms) but complementary, flows of information.

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